



DEPARTMENT OF COMPUTER SCIENCE

COMP4381 - Data Science and Analytics

Analyzing and Predicting Flight Delays Around the World

Prepared by:

Cileen Antari - 1220944

Shaymaa Dar Mousa - 1222002

Shahd Alhih - 1221817

Instructor's Name: Dr. Ahmed Sabbah

Section: 2

Date: 27/4/2026

Abstract

This project aims to study and predict flight delays using simple machine learning techniques. Flight delays are a common problem that affects both passengers and airlines, so understanding them is important. In this project, we will use a large flight dataset from Kaggle as the main source of data. We will also use airport data from the OurAirports dataset to add more useful information. These datasets will be combined to improve the analysis. The goal is to build a model that can predict whether a flight will be delayed or not. We will use a Decision Tree algorithm because it is simple and easy to understand. The results of this project will help us understand the main reasons behind flight delays.

Table of Contents

1. Introduction.....	3
2. Background	3
3. Domain Overview.....	4
4. Algorithm Overview	5
4.1. How a Decision Tree Works.....	5
4.2. Advantages of Decision Trees.....	5
5. Literature Review	6
5.1. Paper 1.....	6
5.2. Paper 2.....	6
5.3. Paper 3.....	7
5.4. Paper 4.....	8
5.5. Paper 5.....	8
6. References.....	10

1. Introduction

The aviation industry plays an important role in connecting countries and supporting travel, business, and tourism around the world. Every day, a large number of people depend on flights to reach their destinations. However, flight delays are a common problem that can affect passengers and airlines in many ways.

Flight delays can happen for different reasons such as bad weather, airport congestion, technical problems, or delays from previous flights. These delays can cause inconvenience for passengers, missed connections, and additional costs for airlines. Because of this, it is important to understand the main factors that lead to delays and try to predict them.

With the help of data science, it has become possible to analyze large amounts of flight data and discover patterns related to delays. In this project, we will use flight data to study the factors that may affect delays and build a simple model to predict whether a flight will be delayed or not. The goal is to better understand this problem and provide useful insights that can help improve flight planning and decision-making.

2. Background

Flight delays are one of the common challenges in the aviation industry, affecting both airlines and passengers. When flights are delayed, passengers may experience longer travel times, missed connections, and inconvenience. At the same time, airlines may face higher operational costs and scheduling problems. According to the Federal Aviation Administration (FAA), flight delays cost the U.S. aviation industry around 33 billion dollars every year (FAA, 2021).

Flights can be delayed for several reasons. These include bad weather conditions, heavy traffic at airports, technical problems with aircraft, or issues related to the flight crew. In some cases, delays happen because the same aircraft arrived late from a previous flight. Studies have shown that weather is responsible for a large percentage of flight delays and may account for more than 50% of arrival delays (Mamdouh et al., 2023).

In recent years, the availability of large flight datasets from organizations such as the U.S. Bureau of Transportation Statistics (BTS) and EUROCONTROL has made it possible to analyze and predict flight delays using data science techniques. By combining flight data with airport

information and other related factors, researchers can identify patterns associated with delays and develop prediction models (Dai, 2024).

Machine learning methods have become widely used in this field. Algorithms such as Decision Trees, Random Forests, and Gradient Boosting can analyze historical flight data and predict whether a flight will be delayed or arrive on time. Previous research has also shown that proper data cleaning, feature engineering, and the use of historical delay records play an important role in improving prediction accuracy (AlBassam & AlShahrani, 2025).

3. Domain Overview

Flight delay prediction belongs to the domain of aviation data analytics, where flight operational data is analyzed to understand the factors affecting flight punctuality. The aviation industry plays a key role in global transportation, connecting countries and supporting economic growth. However, flight delays remain one of the most common challenges, negatively impacting both passengers and airline operations (AlBassam & AlShahrani, 2025).

In aviation analytics, a flight is typically considered delayed if it departs or arrives more than 15 minutes after its scheduled time. Delays can occur due to various factors such as weather conditions, airport congestion, technical issues, or delays from previous flights. These delays may lead to missed connections, increased travel time, and higher operational costs for airlines.

The main unit of analysis in this domain is the individual flight record. Each record includes information such as airline, origin and destination airports, scheduled and actual departure and arrival times, and delay-related attributes. In addition, airport datasets provide useful contextual information such as location, airport type, and geographic details. Combining flight and airport data allows for a more comprehensive analysis of delay patterns.

Data science and machine learning play an important role in this domain. By analyzing historical flight data, models can identify patterns associated with delays and predict whether a future flight will be delayed. This helps airlines improve scheduling, optimize resource allocation, and provide more accurate information to passengers. Therefore, flight delay prediction represents a practical and impactful application of data analytics.

4. Algorithm Overview

In this project, machine learning techniques are used to predict whether a flight will be delayed or arrive on time. The primary algorithm applied is the Decision Tree classifier, which is one of the most widely used algorithms for classification problems. Decision Trees belong to the family of supervised learning algorithms, where the model learns patterns from labeled training data. Due to their simplicity, interpretability, and ability to handle different types of data, Decision Trees are well suited for flight delay prediction tasks (AlBassam & AlShahrani, 2025).

4.1. How a Decision Tree Works

A Decision Tree works by recursively splitting the dataset into smaller subsets based on features that best separate the classes. At each node, the algorithm selects the feature that produces the best split using criteria such as Gini Impurity or Information Gain. Gini impurity measures the probability of incorrect classification, while Information Gain measures the reduction in uncertainty after a split (Mitchell, 1997).

The structure of a Decision Tree includes a root node, internal decision nodes, branches representing possible outcomes, and leaf nodes that provide the final prediction. In this project, the predicted classes are “Delayed” and “Not Delayed.” To classify a new flight record, the model starts from the root node and follows the branches until reaching a leaf node.

4.2. Advantages of Decision Trees

Decision Trees offer several advantages that make them suitable for this task. They are easy to interpret, allowing users to understand how predictions are made. They can handle both numerical and categorical data, making them suitable for flight datasets. In addition, they do not require strong assumptions about data distribution and are computationally efficient. They can also identify the most important features affecting flight delays.

Overall, Decision Tree provides a good balance between accuracy, simplicity, and interpretability, making it a suitable choice for this project. Previous studies have shown that it can achieve strong performance in flight delay prediction when combined with proper preprocessing techniques (AlBassam & AlShahrani, 2025).

5. Literature Review

5.1. Paper 1

Flight delay prediction using machine learning and explainable AI: a case study on Hazrat Shahjalal International Airport, Dhaka

Islam et al. (2025) studied flight delay prediction using machine learning and explainable AI at Hazrat Shahjalal International Airport in Dhaka. The study shows that predicting delays is important, but it is also important to understand the reasons behind these delays.

The researchers used flight data together with weather data such as temperature, visibility, and wind speed. This helps in understanding how different factors can affect flight delays. The dataset included several features related to flights, scheduling, and weather conditions.

They tested different machine learning models such as Decision Tree, Random Forest, XGBoost, and CatBoost. The results showed that XGBoost and CatBoost gave the best performance. The study also used LIME to explain how the model makes decisions and which factors affect delays the most. The results showed that weather and scheduling are important factors.

This paper is useful for our project because it helps us understand the main causes of flight delays and supports the use of machine learning models to predict delays.

5.2. Paper 2

Predicting Flight Delays Using Machine Learning

Snell et al. (2026) conducted a study on flight delay prediction using machine learning techniques. The study focuses on improving the accuracy of predicting flight delays and helping airlines manage disruptions more effectively. The researchers used real flight data from the United States for the year 2024, combined with weather data from NOAA.

The dataset included different features such as flight schedules, airport information, and weather conditions like temperature, wind, and humidity. Several machine learning models were tested for both classification and regression tasks. These included Logistic Regression, Decision Tree, k-Nearest Neighbors, Bagging, and Random Forest.

The results showed that ensemble models, especially Random Forest, achieved the best performance with high accuracy. The model was able to detect delayed flights with an accuracy of

about 94.7%. The study also applied SMOTE to handle data imbalance, which improved the performance of the models. In addition, regression models were used to predict the number of delay minutes, showing good accuracy.

This paper is useful for our project because it highlights the importance of combining flight data with weather data and shows that machine learning models, especially ensemble methods, can provide accurate predictions of flight delays.

5.3. Paper 3

Flight delay prediction: Evaluating machine learning algorithms for enhanced accuracy

AlBassam and AlShahrani (2025) presented a study titled “Flight Delay Prediction: Evaluating Machine Learning Algorithms for Enhanced Accuracy,” which focused on improving flight delay prediction using machine learning techniques. The study explained that flight delays are a major problem in the aviation industry because they increase costs, affect airline operations, and reduce passenger satisfaction. Accurate prediction of delays is important for improving scheduling and customer service (AlBassam & AlShahrani, 2025).

The researchers used a large Kaggle dataset called “Flight Status Prediction,” which contains about 4 million records and 61 features from 2018 to 2022. The dataset includes information such as airports, flight numbers, airline codes, scheduled and actual departure and arrival times, cancellations, weather delays, and flight distance (AlBassam & AlShahrani, 2025).

The main goal was to build a model that could classify delays into three levels: small (less than 15 minutes), medium (between 15 and 45 minutes), and large delays (more than 45 minutes). To solve the problem of class imbalance, they applied techniques such as Random Oversampling, SMOTE, and ADASYN (AlBassam & AlShahrani, 2025).

The study tested six machine learning algorithms, including Decision Tree, Random Forest, Logistic Regression, SVC, KNN, and Naive Bayes. The results showed that Random Forest with Random Oversampling and Decision Tree with SMOTE gave the best performance, reaching an accuracy of 0.90 and an F1-score of 0.91. Decision Tree was especially useful because it was accurate and easy to understand.

This paper is important for our project because it supports using classification models for flight delay prediction. It also confirms that Decision Tree is a strong and suitable algorithm for predicting whether a flight will be delayed or not, which matches the main goal of our project.

5.4. Paper 4

A hybrid machine learning-based model for predicting flight delay through aviation big data

Dai (2024) proposed a hybrid machine learning framework for predicting flight delays using aviation big data collected through web crawling from FlightRadar24, focusing on flights arriving at John F. Kennedy International Airport during 2023. The dataset consisted of 24,986 flight records described by 36 features related to weather conditions, flight characteristics, and airport congestion. To identify the most relevant predictors, the study employed a combination of one-way ANOVA and Forward Sequential Feature Selection (FSFS), reducing the feature set to 21 statistically significant variables. The analysis indicated that weather conditions, airport congestion levels, and historical delay records were the most influential factors affecting flight delays. An enhanced DBSCAN clustering algorithm was then applied to partition the training data into homogeneous groups, enabling separate models to be trained for each cluster and improving both computational efficiency and prediction accuracy.

Building on this pipeline, the author introduced a novel ensemble model called COWRF (COA-optimized Weighted Random Forest), where the Coyote Optimization Algorithm optimizes decision tree split points and assigns weighted voting scores to each tree. Using 50-fold stratified cross-validation, the proposed model achieved an accuracy of 97.2%, outperforming several baseline models including conventional Random Forest (91.90%), CNN-LSTM (91.10%), and DFCNN (92.04%). These findings provide valuable insights for our project, particularly in guiding feature selection strategies, preprocessing methods, and the adoption of ensemble learning models as strong baselines for performance comparison.

5.5. Paper 5

A novel intelligent approach for flight delay prediction. *Journal of Big Data*

Mamdouh et al. (2023) proposed a flight delay prediction framework called FDPP-ML that relies only on basic flight schedule features such as origin, destination, carrier, flight number, and scheduled departure and arrival times. The dataset was obtained from the Bureau of Transportation Statistics and included more than 5.5 million flight records covering 366 airports and 10 airlines in the United States between May and December 2019. The authors observed a strong correlation between departure and arrival delays and therefore introduced two engineered features: Previous Flight Delay (PFD) and Flight Time Duration (FDT). The proposed framework restructures flight data into sequential path-based points and propagates delay information along each flight path to

improve prediction capability. The model was evaluated using several machine learning algorithms, including Random Forest, LightGBM, CatBoost, GRU, and LSTM, where the Stacking ensemble model achieved the best overall performance. Experimental results showed that the proposed approach reduced prediction errors by up to 32% in Mean Absolute Error (MAE) and 38% in Mean Squared Error (MSE) at a 2-hour prediction horizon compared with traditional models. These findings are relevant to our project as they highlight the importance of historical delay propagation as an engineered feature and demonstrate that competitive prediction performance can be achieved using publicly available schedule data without requiring complex weather or airport-specific features.

Overall, previous studies have shown that machine learning techniques are effective in predicting flight delays. Different models such as Decision Trees, Random Forests, and ensemble methods have achieved strong performance. These studies also highlight the importance of data preprocessing and feature selection. Based on this, our project focuses on using a Decision Tree model to build a simple and interpretable prediction system.

6. References

1. AlBassam, S. A. A., & AlShahrani, D. N. (2025). Flight delay prediction: Evaluating machine learning algorithms for enhanced accuracy. *PLOS One*, 20(12), e0335141. <https://doi.org/10.1371/journal.pone.0335141>
2. Mitchell, T. M. (1997). *Machine Learning*. McGraw-Hill. [Open Machine Learning Book](#)
3. Dai, M. (2024). A hybrid machine learning-based model for predicting flight delay through aviation big data. *Scientific Reports*, 14, Article 4603. <https://doi.org/10.1038/s41598-024-55217-z>
4. Mamdouh, M., Ezzat, M., & Hefny, H. A. (2023). A novel intelligent approach for flight delay prediction. *Journal of Big Data*, 10, Article 179. <https://doi.org/10.1186/s40537-023-00854-w>
5. Federal Aviation Administration. (2021). *Aviation data and statistics*. <https://www.faa.gov/dataresearch/aviationdatastatistics.pdf>
6. Islam, M. R., et al. (2025). Flight delay prediction using machine learning and explainable AI: A case study on Hazrat Shahjalal International Airport, Dhaka. *Journal of Air Transport Management*. <https://doi.org/10.1016/j.jairtraman.2025.103083>
7. Snell, K., Zurada, J., Kozak, J., Hatami, Z., & Olszak, C. (2026). Predicting flight delays using machine learning. *Proceedings of the 59th Hawaii International Conference on System Sciences*. <https://scholarspace.manoa.hawaii.edu/server/api/core/bitstreams/ed6857c4-0adf-4019-9f8c-60378561de0e/content>
8. Khaksar, H., & Sheikholeslami, A. (2017). Airline delay prediction by machine learning algorithms. *Scientia Iranica*, 26(5). <https://doi.org/10.24200/sci.2017.20020>